A Posteriori Error Certificates (APECs): Certifying Uncertainty Through the Soft-Mode Bottleneck in $BaTiO_3$

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Electronic structure theory has become an indispensable tool for predicting functional materials properties, yet computational predictions are rarely accompanied by standardized uncertainty estimates. This limitation is particularly important for derived materials properties, where nonlinear relationships can amplify uncertainties in the underlying electronic structure calculation. Here we introduce A Posteriori Error Certificates (APECs), a method that attaches quantitative uncertainty estimates, provenance metadata and quality tiers to density functional theory (DFT) calculations. Total uncertainty is decomposed into convergence, methodological, intrinsic DFT and model-form contributions, and propagated through derived quantities to enable statistically consistent fusion of heterogeneous computational data. We demonstrate the framework for the electro-optic response of tetragonal $BaTiO_3$, where the inverse-square dependence of the Pockels coefficient on the soft-mode frequency strongly amplifies uncertainties in the underlying phonon calculation. Twelve certificates are extracted from a production density-functional perturbation theory calculation and combined with cross-functional literature data. Phase-aware partitioning resolves inconsistencies arising from mixed crystallographic phases and tensor conventions, yielding a fused electro-optic coefficient within 3% of the bulk experimental value while reducing the uncertainty from 28% for a single calculation to approximately 24% through principled data fusion. More broadly, APECs provide a methodology for uncertainty-aware computational materials science, enabling calculated properties to be compared, combined, propagated through multistage workflows, and reused reliably.

I. INTRODUCTION

Density functional theory plays a central role in computational materials science, enabling prediction of materials properties and the discovery of new functional materials [1–3]. Despite its widespread use, computational predictions are rarely accompanied by standardized quantitative uncertainty estimates. This limitation is particularly important for derived materials properties, where nonlinear relationships can amplify uncertainties in the underlying electronic structure calculations. Reliable uncertainty estimates are thus essential for comparing results from different computational approaches, combining heterogeneous datasets, and supporting data-driven materials discovery [4–6].

Barium titanate ($BaTiO_3$) provides a particularly demanding example of this problem. It is among the most promising materials for integrated electro-optical modulators because of its large Pockels coefficients and compatibility with silicon photonic platforms [7–9]. Predicting the electro-optic response from DFT is therefore a problem of direct technological significance. It is also a problem that exposes a fundamental weakness in how computational materials science handles uncertainty: standard methodological errors in DFT are often amplified when calculating derived properties.

The Pockels coefficient r_{33} of $BaTiO_3$ depends on the soft transverse optical phonon frequency ω_{soft} through an inverse-square relationship [10],

$$r_{33} \propto \frac{Z^{*2}}{\omega_{\mathrm{soft}}^2} \tag{1}$$

where Z^* is the Born effective charge. A modest 5% uncertainty in ω_{soft}, well within the variation among standard exchange-correlation functionals, becomes a 10% uncertainty in r_{33}. The published DFT values of r_{33} for bulk $BaTiO_3$ vary by a factor of two or more depending on the functional [10–12], reflecting the increased sensitivity to ω_{soft}. Near the Curie temperature, where $\omega_{\mathrm{soft}} \to 0$, this amplification diverges entirely.

This amplification is not unique to $BaTiO_3$ Pockels coefficients or ferroelectrics. Any property with nonlinear dependence on DFT outputs will behave similarly: piezoelectric coefficients depend on phonon frequencies and Born charges [13], defect formation energies enter Arrhenius rates exponentially [14], and catalytic turnover frequencies depend exponentially on adsorption energies [15, 16]. In each case, modest input uncertainties propagate into large output uncertainties through the physics of the problem. Our group encountered similar challenges in electronic structure calculations of response properties in heterogeneous nanocomposites, where subtle changes in electronic structure, inclusion morphology, and interfacial polarization can produce significant changes in the predicted material response [17, 18].

Yet, no standardized quantification of uncertainty in DFT calculations exists [19]. Predicted properties of-

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ten vary widely with the choice of exchange-correlation functionals, basis sets and convergence parameters. Of particular relevance for materials calculations are band gaps. These are often underestimated by LDA and GGA functionals by $35-40\%$, whereas hybrid functionals vary by $\pm 15\%$ depending on the proportion of exact exchange included [20, 21]. Even lattice constants differ systematically across functionals with variations of the order of 0.5% [22, 23], whereas formation energies for transition metal oxides show spreads of $0.1-0.3$ eV[24, 25], and elastic constants show variations of $10-30\%$ for complex materials [26].

This heterogeneity becomes a systemic problem as computational materials science scales. Large computational materials databases (Materials Project, AFLOW, JARVIS) aggregate millions of entries generated using different computational approximations [1–3, 27–29]. These results are often treated on equal footing in data-driven approaches such as machine learning, despite potentially significant differences in reliability and systematic errors or biases [30].

Several prior approaches have been devised to address DFT uncertainty. The Δ-project [22] and subsequent automated convergence workflows [31] established that different DFT codes agree to within ~ 1 meV/atom when using identical parameters, demonstrating that *implementation* differences are negligible and that the dominant uncertainty source is the choice of method, not the choice of code. The BEEF-vdW functional ensemble [4, 32] estimates methodological uncertainty directly through Bayesian sampling of exchange-correlation functionals, but is restricted to the BEEF family and cannot be applied retrospectively to existing datasets. Statistical approaches have been developed to estimate predictive uncertainty from benchmark datasets, including systematic calibration models by Pernot et al. [5, 6, 33] and uncertainty-aware probability distributions over exchange-correlation functional space [34]. These approaches provide an empirical basis for uncertainty estimates, but not per-calculation uncertainty representation. At the database level, ad hoc correction schemes such as the GGA/GGA+U mixing for transition metal oxides [35–37] improve accuracy but do not provide uncertainty estimates for individual entries. Finally, automated convergence frameworks [38–41] reduce the cost of numerical validation but yield pass/fail criteria rather than quantitative uncertainty measures.

The first-principles electro-optic tensor is not, in principle, inaccessible. Density-functional perturbation theory (DFPT) and related finite-field approaches can compute electronic, ionic, and piezoelectric contributions to the Pockels response directly for ideal bulk crystals [10, 42]. The challenge addressed here is different. Many technologically relevant systems involve strain, defects, interfaces, domains, finite-temperature disorder or large low-symmetry supercells, making repeated full-tensor calculations computationally demanding. In these systems, a calibrated soft-mode description provides a compact, physically interpretable descriptor whose assumptions, calibration and uncertainty amplification can be certified explicitly.

Here, we introduce a methodology that treats uncertainty as structured metadata attached to computational materials data. Rather than being tied to a particular exchange-correlation functional, workflow or dataset, the resulting A Posteriori Error Certificates (APECs) can be generated retrospectively for completed calculations and incorporated into existing databases. Each computed quantity of interest is provided with an uncertainty certificate. The certificate decomposes uncertainty into convergence, methodological, intrinsic DFT and model-form contributions, records computational provenance, and assigns a quality tier.

The certificate framework also enables uncertainty propagation through derived quantities and statistically consistent combination of heterogeneous data. Because the individual sources of uncertainty are retained explicitly rather than merged into a single confidence estimate, certificates generated for individual calculations can be compared and combined across computational approaches while preserving information about the origin of the uncertainty. The framework further identifies which uncertainty contributions dominate a given prediction, helping prioritize future methodological developments, additional benchmark data, or more stringent convergence studies. Together, these capabilities extend uncertainty quantification beyond individual calculations, allowing computational materials data to be compared, combined, propagated through multistage workflows, and reused through a systematic treatment of uncertainty.

We apply the methodology to the electro-optic response of $BaTiO_3$, using the bulk soft-mode response as a tractable proxy for a full tensor quantity that becomes costly to evaluate repeatedly in large or low-symmetry configurations. In this system, soft-mode amplification of errors makes uncertainty quantification essential. Using a reference DFPT calculation, we extract twelve certificates spanning structural, charge-response, dielectric, phonon, and derived electro-optic properties. We demonstrate phase-aware fusion of heterogeneous data sources and quantify the resulting reduction in uncertainty. We assign to these values an expected quality tier: green for well-converged ground-state properties, amber for response properties, and red for the derived Pockels coefficient subject to both propagated uncertainty and model-form error. The $BaTiO_3$ case study serves as a controlled demonstration of the certificate framework and illustrates how uncertainty-aware data aggregation can enhance the reliability of computational predictions of materials properties.

II. RESULTS

The APECs method assigns uncertainty certificates to individual DFT-calculated and derived quantities, and enables their combination across heterogeneous data sources. We demonstrate the workflow using the calibrated soft-mode electro-optic response of $BaTiO_3$, where a compact descriptor stands in for a full tensor calculation within a clearly stated bulk reference scope. We build the certificates from calculated, literature and database data, quantifying the dominant uncertainty contributions. Specifically, in the following pages we generate certificates for a production DFPT calculation and examine how library fusion and provenance-aware partitioning affect the resulting electro-optic effect.

In $BaTiO_3$, a transverse optical phonon mode softens as temperature approaches the Curie point $T_C \approx 400$ K [9, 43–45]. The change in the refractive index of the material with an applied electric field is determined by the Pockels coefficient r_{33} which depends on this soft-mode frequency as discussed in Equation 1. More specifically [10],

$$r_{33} = r_{\text{ref}} \times \left(\frac{\omega_{\text{ref}}}{\omega}\right)^2 \times \left(\frac{Z^*}{Z^*_{\text{ref}}}\right)^2 \times \left(\frac{\varepsilon_{\text{ref}}}{\varepsilon^{\infty}}\right). \quad (2)$$

Here, Z^* is the Born effective charge associated with the soft polar mode [46–48]. Reference values are the bulk experimental value of Zgonik et al. [49] $r_{\text{ref}} = 105$ pm/V, $\omega_{\text{ref}} = 180$ cm^{-1} [9], $Z^*_{\text{ref}} = 7.16\ e$ [46], and $\varepsilon_{\text{ref}} = 6.5$ [46].

Uncertainty in the quantities entering Equation 2 propagates into the derived electro-optic response according to:

$$\left(\frac{\sigma_{r_{33}}}{r_{33}}\right)^2 = \left(2\frac{\sigma_\omega}{\omega_{\text{soft}}}\right)^2 + \left(2\frac{\sigma_{Z^*}}{Z^*}\right)^2 + \left(\frac{\sigma_\varepsilon}{\varepsilon_\infty}\right)^2. \quad (3)$$

Each quantity appearing in Equation 2 is associated with an APEC, a structured metadata record that attaches quantitative uncertainty estimates to DFT-computed and derived properties. The method decomposes uncertainty for each property (such as the ones in Equation 3) into convergence uncertainty (σ_{conv}), methodological uncertainty (σ_{method}), intrinsic DFT uncertainty (σ_{DFT}), and, for derived quantities, model-form uncertainty (σ_{model}):

$$\sigma_{\text{total}} = \sqrt{\sigma^2_{\text{conv}} + \sigma^2_{\text{method}} + \sigma^2_{\text{DFT}} + \sigma^2_{\text{model}}}, \quad (4)$$

where σ_{total} denotes the total uncertainty associated with a given observable, such as $\sigma_{r_{33}}$, σ_ω, σ_{Z^*}, or σ_ε.

These uncertainties are reported together with the computed values and provenance metadata, can be added to existing data, and are classified into discrete quality tiers (Red/Amber/Green) that facilitate comparison across calculations and data sources. The certificates further provide a mechanism for propagating uncertainty through derived quantities and for combining information from heterogeneous literature and database sources in a consistent manner.

A. Extracting Certificates from Literature and Database Sources

To construct APECs for $BaTiO_3$, literature values and computational data were collected from published studies and public materials databases. We developed workflows for data extraction and retroactive uncertainty estimation for three source types: a literature compilation from published DFPT studies spanning 1994–2025 [50, 51], Materials Project entries (mp-5986 for the tetragonal phase, mp-2998 for cubic) with automated uncertainty estimation, and the Kyoto PhononDB with systematic phonon calculations [52, 53]. Most DFT data in these public repositories is reported without uncertainty estimation. The present workflow constructs APECs from the reported computational settings, convergence behaviour and cross-functional comparisons where available.

Public databases also typically lack explicit convergence uncertainty information. Our method applies a hierarchy of increasingly conservative estimation strategies to address this. When cutoff and k-point parameters are reported but convergence tests are unavailable, we assign representative convergence estimates (see Tables S3–S4) using typical convergence behavior in perovskite DFPT calculations, informed by published convergence studies [11, 54] and cross-checked against our own $BaTiO_3$ calculations. These estimated ranges are provisional rather than universal, and should be refined as systematic convergence benchmarks become available. When calculations of the same property are available for multiple functionals, the cross-functional spread provides a direct estimate of σ_{method} via Equation 12. Missing metadata automatically triggers quality tier downgrading from Green to Amber or Red. All values carry provenance metadata, indicating whether the uncertainty is derived from a convergence test, a literature range, or a heuristic estimate.

The final quantitative electro-optic posterior reported below is driven by the curated DFPT literature library, typed PhononDB support where the target observable and phase are clear, and the production Quantum ESPRESSO calculation. Materials Project and other public database records are retained primarily as retrospective extraction and metadata-stress tests unless the phase, target observable, and tensor convention are sufficiently typed for exchangeable fusion. This policy prevents incompletely typed database records from dominating the fused electro-optic estimate.

The compiled literature library serves as an empirical reference distribution against which new $BaTiO_3$ calculations can be compared and cross-validated. By aggregating values from multiple groups, codes, and functionals over three decades of DFPT work, the data capture the empirical spread that defines methodological uncertainty and provides the multi-source inputs needed for inverse-variance weighted averaging. Table I summarizes these data, including recent cross-functional comparisons [55, 56] and our own production calculations. The library includes entries spanning three exchange-correlation functionals (LDA, PBE, PBEsol), and experimental references, along with experimental Pockels coefficient measurements from bulk crystals and epitaxial thin films [10, 44, 46, 57–60]. In particular, the Z^*_{Ti} and ε^∞ entries illustrate the importance of phase-aware partitioning: older cubic-reference or high-symmetry reference-structure values differ significantly from explicitly tetragonal values for the same descriptor.

B. Convergence Analysis

Different uncertainty sources dominate different observables in $BaTiO_3$. Structural quantities converge relatively rapidly with respect to cutoff energy and k-point sampling, whereas response properties remain substantially more sensitive to numerical and methodological choices [61]. Convergence analysis helps identify which computational parameters limit the accuracy for each quantity entering the calculation of the Pockels coefficient, and where additional computational investment will or will not reduce uncertainty.

Soft-Mode Frequency. The soft-mode frequency is the most convergence-sensitive input to the Pockels coefficient, requiring significantly higher cutoff energies and denser k-point meshes than structural properties. Figure 1 shows heuristic response-property convergence estimates for ω_{soft} based on the ranges summarized in Table S4, shown as visual interpolants rather than as explicit DFPT convergence calculations at every point. Only the starred points correspond to evaluations at the production-calculation parameters; the connecting curves are guides constrained to reproduce the class-level ranges of Table S4 and are not point-by-point DFPT results. The soft-mode frequency converges more slowly than total energy or forces, requiring $E_{\mathrm{cut}} > 80$ Ry for $< 2\%$ error and a k-point mesh of $8 \times 8 \times 8$ or denser. At the parameters used in our production calculation (60 Ry, $6\times6\times6$), the estimated convergence uncertainty is $\sigma_\omega/\omega \approx 3\%$, consistent with the heuristic estimates in Table S4.

Born Effective Charges. Born effective charges converge more rapidly than the soft mode. At 60 Ry, $\sigma_{Z^*}/Z^* < 1\%$ is already achieved, with relatively weak dependence on k-point sampling. Unlike the soft-mode frequency, the Born charges therefore contribute only modestly to the convergence component of the final r_{33} certificate.

Propagated Uncertainty in r_{33}. The key question is how convergence uncertainties in the individual inputs propagate into the derived Pockels coefficient. Applying Equation 3 with $\sigma_\omega/\omega \approx 3\%$ and $\sigma_{Z^*}/Z^* \approx 1\%$ yields a propagated uncertainty of approximately 6.3%.

The soft-mode convergence dominates the propagated uncertainty, as expected from the sensitivity analysis. Including ε_∞ convergence from Equation 3 (1.5%, entering with exponent 1) gives $\sigma_{\mathrm{conv}} = 6.5\%$ for r_{33}.

C. Independent SCF Convergence Study

To validate the convergence estimates against explicit calculations, we performed an independent Quantum ESPRESSO SCF convergence study for the same 40-atom tetragonal $BaTiO_3$ supercell used in the certificate workflow, shown in Figure 2. The study includes 37 single-point calculations using the PBE and PBEsol functionals, with systematic variation of both plane-wave cutoff and Monkhorst-Pack k-point sampling. All runs converged to SCF residuals below 10^{-10} Ry, and the highest-cutoff/highest-k case within each functional is used as the internal reference for the corresponding functional.

The convergence study shows that the total energy for this supercell controlled primarily by the plane-wave cutoff rather than by k-point sampling. For both PBE and PBEsol, the production-like 60 Ry, $6 \times 6 \times 6$ calculations remain approximately $10 - 11$ meV/atom above the highest-cutoff reference calculations, while increasing the cutoff to 80 Ry reduces the residual error below 1 meV/atom across all sampled k-point meshes. By contrast, the residual variation among k-point meshes at the maximum cutoff is negligible (roughly 10^{-3} meV/atom for both PBE and PBEsol). This SCF campaign certifies the numerical scale of total-energy convergence for the 40-atom supercell only. It does not replace property-specific DFPT convergence tests for soft phonons, dielectric response, Born charges, or electro-optic tensors. The residual 60 Ry total-energy offset therefore reinforces, rather than relaxes, the conservative response-property convergence estimates used in the certificates. These results support the use of conservative cutoff-driven convergence uncertainties for structural and total-energy quantities, while also illustrating why response-property convergence must be treated separately: soft phonons and electro-optic tensors can remain sensitive even after the ground-state energy is numerically stable.

D. Methodological Spread

Even after numerical convergence is achieved, substantial uncertainty remains due to the choice of exchange-correlation functional. For response properties in $BaTiO_3$, these biases are the dominant uncertainty source and must be estimated from the cross-functional spread in published calculations.

Table S2 compiles published soft-mode frequencies for tetragonal $BaTiO_3$ across LDA, PBE and PBEsol calculations. The cross-functional spread in ω_{soft} itself is relatively modest ($\sigma \approx 5$ cm^{-1}, or $\sim$3%). The production certificate treats this measured spread together with a conservative mode-extraction and response-setup allowance as an 8% soft-mode methodological prior; the impact of this allowance on the final r_{33} certificate is

TABLE I. $BaTiO_3$ library summary. Best estimate values are weighted averages within each reported subset. The cubic Born-charge row groups older cubic-reference or high-symmetry reference-structure entries, for which tensor directions are equivalent by symmetry; the tetragonal zz row contains explicitly tetragonal zz records.

Parameter	Phase	Sources	Functionals	Best estimate
Z^*_{Ti} (isotropic)	Cubic ref.	4 DFT	LDA, PBE	7.27 ± 0.17
$Z^*_{\mathrm{Ti},zz}$	Tetragonal	1 DFT	PBE	6.20 ± 0.20
ε^∞ (avg)	Tetragonal	3 DFT	LDA, PBE	6.77 ± 0.14
ε^∞_{zz}	Tetragonal	1 DFT	PBE	5.96 ± 0.18
ω_{soft}	Tetragonal	5 DFT	LDA, PBE, PBEsol	176 ± 6 cm^{-1}
P_s	Tetragonal	3 DFT, 1 Expt	LDA, PBE, PBEsol	0.27 ± 0.02 C/m^2
c/a	Tetragonal	3 DFT, 1 Expt	LDA, PBE, PBEsol	1.011 ± 0.003
a (lattice)	Tetragonal	4 DFT, 1 Expt	LDA, PBE, PBEsol	3.993 ± 0.015 Å

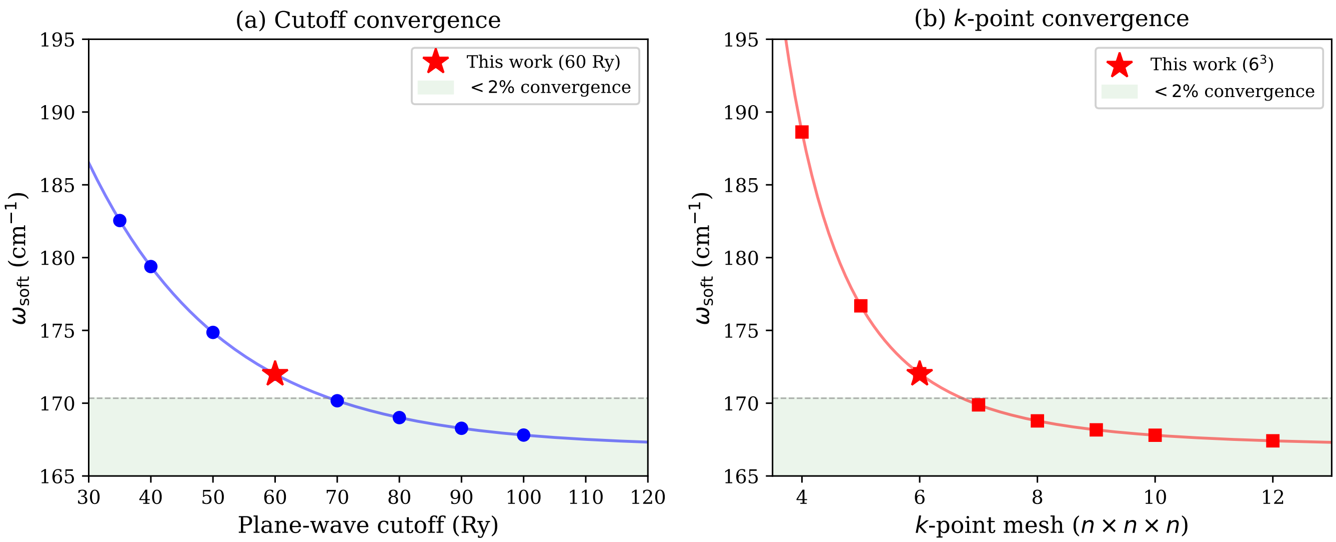


FIG. 1. Heuristic convergence estimates for the soft-mode frequency in $BaTiO_3$. Curves are visual interpolants constrained to reproduce the class-level DFPT uncertainty ranges summarized in Table S4; they are not explicit DFPT convergence calculations at every point and are not fitted from the preliminary transferable convergence parameters in Table S3. Only the starred points correspond to evaluations at the production-calculation parameters (PBE/PAW, 60 Ry, 6^3 k-points). The green band indicates the $< 2\%$ convergence region.

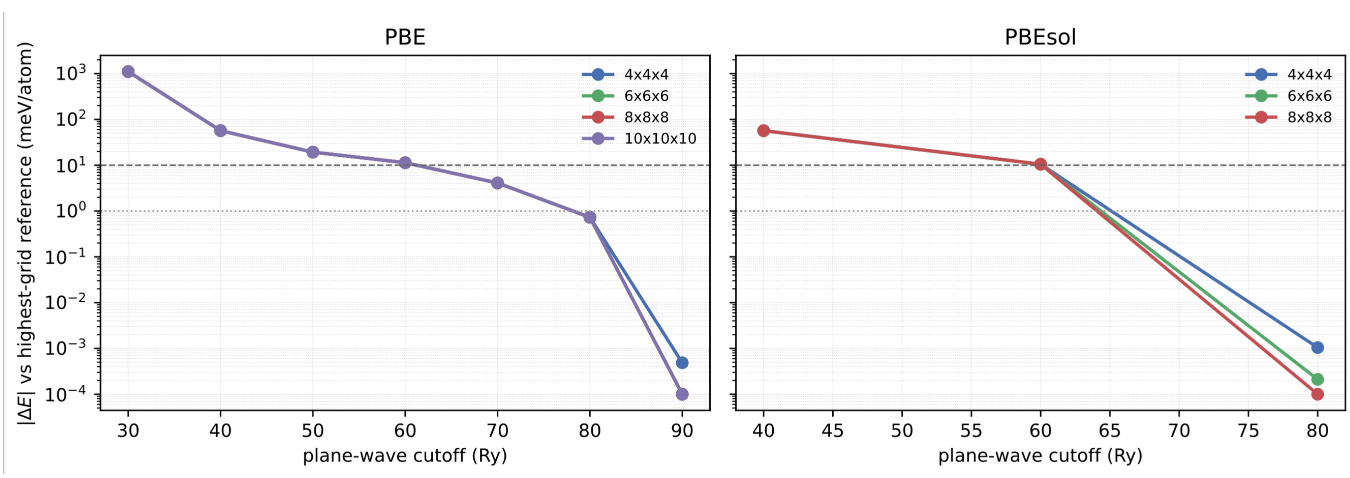


FIG. 2. SCF total-energy convergence study for the 40-atom tetragonal $BaTiO_3$ supercell. Energies are shown relative to the highest-cutoff, densest-k-point calculations performed for each functional (PBE: 90 Ry, $10 \times 10 \times 10$; PBEsol: 80 Ry, $8 \times 8 \times 8$). The dashed and dotted horizontal lines mark 10 and 1 meV/atom thresholds. Energy convergence is dominated by the plane-wave cutoff, whereas the residual k-point dependence at the highest cutoff is below 2×10^{-3} meV/atom.

audited below. Born effective charges show larger functional sensitivity: published Z^*_{Ti} values range from 7.0 to 7.5 e (a spread of ∼4%) across the available DFPT studies [10, 44, 46, 62]. When propagated onto the soft-mode expression for r_{33} using the certificate line items, these variations produce a methodological uncertainty of 17.8%, substantially larger than the corresponding numerical convergence contribution. This is consistent with the full uncertainty decomposition shown in Figure 3.

E. Certificate Generation

With the convergence and methodological components characterized, we can now assemble complete certificates for all 12 properties from a single DFPT calculation. Figure 3 shows the 12 certificates for the $2 \times 2 \times 2$ tetragonal $BaTiO_3$ supercell calculation (PBE/PAW DFPT Quantum ESPRESSO 7.5 [63], 60 Ry cutoff, $6 \times 6 \times 6$ k-point mesh). The complete four-component uncertainty decomposition for all twelve certificates is tabulated in Table S7. The certified quantities span structural, dielectric, phonon, and derived electro-optic properties, together with their uncertainty decomposition and quality tier assignment.

We distinguish three levels of certification. Layer 1 consists of direct production DFPT certificates for structural, dielectric, phonon, and charge-response descriptors. Layer 2 propagates these descriptor certificates through the calibrated soft-mode model to obtain derived r_{33} certificates. Layer 3 fuses exchangeable literature, database, and production certificates after partitioning by phase, tensor component, and target observable. Only Layer 3 values are library-fused posteriors; Layer 2 values are single-calculation derived certificates.

For the 10 directly computed properties, $\sigma_{\mathrm{model}} = 0$ and the certificates contain three active uncertainty components. For the derived electro-optic response, however, all four uncertainty layers contribute significantly. The decomposition for r_{33} includes: $\sigma_{\mathrm{conv}} = 6.5\%$ $\sigma_{\mathrm{DFT}} = 11.2\%$; $\sigma_{\mathrm{method}} = 17.8\%$; and $\sigma_{\mathrm{model}} = 17.9\%$, combining to a total relative uncertainty of 28.4% and placing the derived Pockels coefficient in the Red tier.

The model-form contribution in the above reflects the limitations of the soft-mode approximation itself together with uncertainty in the reference calibration parameters. Veithen et al. [10] found that the soft-mode model reproduced approximately 80–85% of the ionic electro-optic response in $BaTiO_3$, with the remaining discrepancy arising from electronic and multi-mode contributions. We adopt a nominal 15% soft-mode approximation uncertainty and combine it with the 9.8% calibration uncertainty to obtain $\sigma_{\mathrm{model}} = 17.9\%$. Similar soft-mode dominance has also been reported for $LiNbO_3$[42], whereas $PbTiO_3$ shows dominant electronic contributions instead.

The resulting quality tier distribution (6 Green certificates, 4 Amber, 2 Red) is typical for well-converged DFPT calculations: structural and charge-response properties converge well, dielectric and phonon properties retain larger methodological sensitivity, and derived electro-optic quantities inherit amplified uncertainty through the soft-mode dependence.

F. Library Update: Incorporating 2024–2025 Data

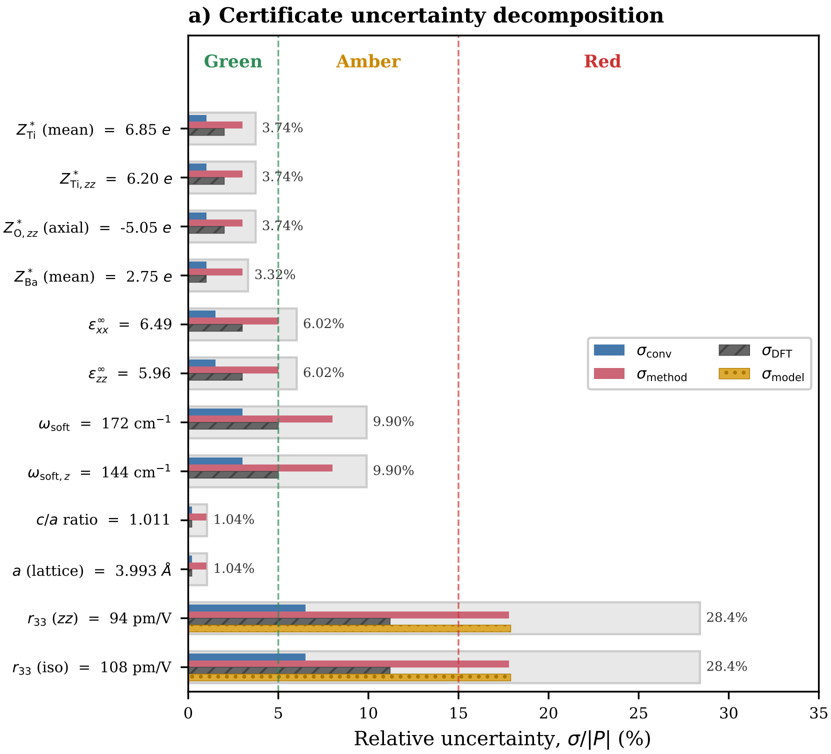
To test how single-calculation APECs fuse together as new calculations become available, we consider six additional entries from recent DFPT studies and compute the resulting fused inverse-variance-weighted error estimates. PBEsol and PBE entries were included from the study by Amounas et al. [55] testing 24 exchange-correlation functionals in Quantum ESPRESSO for the spontaneous polarization and refractive indices of tetragonal $BaTiO_3$. GGA+U and hybrid functional optical properties at room temperature from the work by Inerbaev et al. [56], providing additional cross-checks on the dielectric response, were also included. Combined with our own DFPT results from Figure 3, these additions increase the library from 25 to 31 entries.

Building on the phase-aware partitioning introduced in Table I, Table II quantifies its effect on fused library averages and derived r_{33}. The unpartitioned library conflates cubic-reference and tetragonal-phase values for Z^*_{Ti} and mixes isotropic averages with zz components for ε^{∞}. When the tetragonal-phase PBE values are added without partitioning, the uncertainty in ε^{∞} *widens* by 60%, indicating a data curation problem, not a solution to it. Partitioning by phase and tensor component resolves the discrepancies and yields consistent fused estimates.

The impact on the derived r_{33} is substantial. The unpartitioned library average ($r_{33} = 114 \pm 27$ pm/V) is inflated primarily because it draws Z^*_{Ti} predominantly from cubic-reference and high-symmetry reference-structure DFPT records [10, 44, 46] where the equivalent tensor-component average is $Z^* = 7.29\ e$. In the explicitly tetragonal phase, the corresponding component drops to $6.20\ e$ due to the ferroelectric distortion. Since $r_{33} \propto Z^{*2}$, this difference alone increases the fused electro-optic response by $(7.29/6.85)^2 \approx 1.13$.

When tetragonal-phase isotropic averages are used consistently ($Z^*_{\mathrm{Ti}} = 6.85\ e$, $\varepsilon^{\infty} = 6.31$, $\omega_{\mathrm{soft}} = 172\ \mathrm{cm}^{-1}$), the model gives $r_{33} = 108$ pm/V, close to the bulk reference value of Zgonik et al. (105 ± 15 pm/V). Because the soft-mode expression uses that bulk value as its calibration anchor, this agreement is an internal consistency check of the phase/tensor partitioning and descriptor extraction, not an independent validation of the absolute electro-optic scale. Since the experimental measurement provides an averaged bulk response, comparison with the isotropic convention is more appropriate than comparison with a single tensor component. The zz-component value ($r_{33} = 94$ pm/V) is instead the relevant prediction for a single-domain, z-polarized device geometry.

The single-calculation certificates in Figure 3 are fully consistent with the corresponding fused library values.



b) Example JSON certificate


```
{
  "parameter": "r_33_iso",
  "value": 108.3,
  "unit": "pm/V",
  "uncertainties": {
    "convergence": 6.5,
    "methodological": 17.8,
    "intrinsic_dft": 11.2,
    "model_form": 17.9,
    "total": 28.4
  },
  "quality_tier": "Red",
  "provenance": {
    "code": "Quantum ESPRESSO",
    "version": "7.5",
    "functional": "PBE",
    "pseudopotential": "PAW",
    "phase": "tetragonal",
    "tensor_component": "all-isotropic",
    "energy_cutoff_Ry": 60,
    "kpoint_mesh": [
      6,
      6,
      6
    ],
    "supercell": "2x2x2",
    "n_atoms": 40
  },
  "calibration": {
    "Z_ref_e": 7.16,
    "epsilon_ref": 6.5,
    "model": "soft-mode r33"
  },
  "schema_version": "posterior-certificate-v1"
}
```


All uncertainty components are relative percentages.
The full table contains one record per certified property.

FIG. 3. Posterior certificates from the $BaTiO_3$ $2 \times 2 \times 2$ supercell DFPT calculation (PBE/PAW, 60 Ry cutoff, $6 \times 6 \times 6$ k-point mesh). (a) Computed structural, dielectric, phonon, and electro-optic properties together with the decomposition of their relative uncertainties into convergence (blue), methodological (red), intrinsic DFT (dark gray), and model-form (gold) contributions. Light gray bars show the total relative uncertainty $\sigma_{\mathrm{total}}/|P|$. Dashed lines at 5% and 15% mark Green/Amber and Amber/Red tier boundaries, respectively. The derived r_{33} is reported in both zz and isotropic tensor conventions, using literature-standard reference values ($Z^*_{\mathrm{ref}} = 7.16\ e$, $\varepsilon_{\mathrm{ref}} = 6.5$); see Section II F for discussion of the convention dependence. (b) Example JSON representation of a posterior certificate, illustrating the core structured metadata schema used to encode computed values, uncertainty decomposition, provenance information and quality-tier assignment. Full records additionally carry `descriptor_selection` and `model_discrepancy` blocks when needed to audit derived or calibrated quantities; further details are provided in Section S2.

The zz-only certificate gives $r_{33} = 94 \pm 27$ pm/V, matching the tetragonal zz library average (94 ± 22 pm/V), whereas the isotropic certificate gives $r_{33} = 108 \pm 31$ pm/V, matching the tetragonal isotropic library average (108 ± 25 pm/V). The consistency between single-calculation and library-fused values validates the extraction pipeline within the soft-mode model. The 14% difference between zz and isotropic values (94 vs 108 pm/V) quantifies the impact of tensor averaging and reinforces the need to track phase and tensor-convention metadata explicitly.

Beyond resolving convention ambiguities, library fusion reduces the uncertainty of the derived electro-optic response. The single calculation certificates carry a total relative uncertainty of 28.4%, whereas fusing with the phase-filtered literature reduces this to approximately 24% under the optimistic assumption that the effective sample size approaches five. This improvement is primarily driven by a reduction in methodological uncertainty as information from multiple functionals and implementations is combined. The absolute uncertainty bars also shrink, from ± 27 to ± 22 pm/V for the zz convention, from ± 31 to ± 25 pm/V for the isotropic convention.

Data fusion does not reduce the model-form contribution ($\sigma_{\mathrm{model}} = 17.9\%$) or the common-mode intrinsic-DFT contribution ($\sigma_{\mathrm{DFT}} = 11.2\%$), which together with convergence set a floor of $\sqrt{6.5^2 + 11.2^2 + 17.9^2} \approx 22\%$ on the achievable r_{33} uncertainty within the present semilocal-DFT, soft-mode approximation. This floor is conditional on two referenced point estimates that are promoted to standard deviations—the nominal 15% soft-mode approximation error and the propagated intrinsic-DFT term—and should be read as a model-form-limited estimate rather than as a sharply determined bound. Reducing it would require a more accurate phenomenological description or full *ab initio* electro-optic tensor calculation. The reduction in uncertainty upon fusion from 28% to approximately 24% illustrates how the certificate framework can be leveraged to not merely *track* uncertainty but actively *reduce* it through principled data combination, while making clear which uncertainty compo-

TABLE II. Effect of phase-aware library partitioning on key parameters and derived r_{33}. "Unpartitioned" averages all entries regardless of phase; "Phase-filtered" averages within consistent (phase, component) subsets. The r_{33} uncertainties include propagated input uncertainties and 17.9% model-form uncertainty from the soft-mode approximation. For fused rows, the methodological component is reduced by the effective number of exchangeable entries, while the intrinsic-DFT and model-form components are held fixed because the library is dominated by shared semilocal-DFT physics. Within the present soft-mode/semilocal-DFT model, the resulting uncertainty cannot fall below the ∼22% floor set by the convergence, intrinsic-DFT, and model-form terms.

Input convention	Z^*_{Ti} (e)	ε^∞	ω_{soft} (cm^{-1})	r_{33} (pm/V)	σ_{rel}
Unpartitioned library	7.29 ± 0.28	6.51 ± 0.40	176 ± 6	114 ± 27	24%
Tetragonal, isotropic avg	6.85 ± 0.20	6.31 ± 0.25	172 ± 6	108 ± 25	24%
Tetragonal, zz only	6.20 ± 0.20	5.96 ± 0.18	172 ± 6	94 ± 22	24%
Single calc, zz (Figure 3)	6.20	5.96	172	94 ± 27	28%
Single calc, iso (Figure 3)	6.85	6.32	172	108 ± 31	28%
Experiment (bulk, 300 K)	—	—	—	105 ± 15	14%

nents cannot be averaged away.

Two sensitivities are useful for interpreting the fused value. First, the 8% methodological prior assigned to the soft-mode input is larger than the measured cross-functional spread of the compiled tetragonal frequencies (∼3%) because it includes a conservative allowance for mode extraction and response-setup ambiguity. If the soft-mode prior were set to 3% or 5%, the single-calculation r_{33} uncertainty would be 24.2% or 25.5%, respectively, rather than 28.4%. Second, the fused uncertainty depends on the effective number of exchangeable sources. Holding the convergence, intrinsic-DFT, and model-form terms fixed, $N_{\mathrm{eff}} = 2$ gives $\sigma_{\mathrm{total}} \approx 25.4\%$, while $N_{\mathrm{eff}} = 5$ gives $\sigma_{\mathrm{total}} \approx 23.5\%$. Thus the rounded "∼24%" value should be read as a high-information, floor-limited case rather than as a universal consequence of adding more entries. A separate descriptor sensitivity is discussed in Section S5.3: substituting the raw z-labeled 144 cm^{-1} mode into the zz soft-mode expression would shift the central estimate from 94 to approximately 134 pm/V. This alternative is not adopted in the reported certificate, but it illustrates why mode provenance is retained explicitly.

A limited leave-one-out calibration test of the phase/component-consistent subsets of the $BaTiO_3$ literature library is presented in Section S3.7. Across 19 residuals in five computational subsets, the pooled residuals give mean $z = -0.03$, RMS $z = 0.99$, reduced $\chi^2 = 0.99$, 68% coverage for $|z| < 1$, and 95% coverage for $|z| < 1.96$. The expected sampling spread of the RMS statistic is approximately $1/\sqrt{2n} = 0.16$, so these values rule out gross miscalibration but do not distinguish fine calibration differences. The pooled value also masks subset-level structure (Table S5): the soft-mode and cubic Z^*_{Ti} subsets are mildly over-dispersed (assigned σ wider than the observed scatter), while the lattice-constant and P_s subsets are mildly under-dispersed, so the pooled RMS near unity reflects partial cancellation of opposite-sign deviations rather than uniform per-subset calibration. Because the standardized residuals are computed against the library's own assigned uncertainties, this test certifies internal consistency between assigned σ values and observed cross-source scatter; it is not an external validation against independent reference data. It is therefore an Amber-level smoke test for gross over- or under-confidence in the $BaTiO_3$ input library, not a broad calibration benchmark across materials or a direct empirical validation of the derived Red-tier r_{33} certificate.

III. DISCUSSION

The results presented in the $BaTiO_3$ case study illustrate several distinct roles of APECs. First, APECs help quantify the uncertainty associated with individual DFT-computed and derived quantities with decomposition into convergence, methodological, intrinsic DFT, and model-form contributions. Second, they enable uncertainty-aware fusion of heterogeneous data sources, allowing literature values, database entries, and new calculations to be combined in a consistent manner. Third, they propagate uncertainty through nonlinear physical relationships, making explicit how uncertainties in computed descriptors influence derived observables. Finally, the uncertainty decomposition helps identify where additional information can most effectively improve predictions. In the present example, numerical convergence contributes relatively little to the total uncertainty of the electro-optic response, whereas methodological and model-form contributions dominate. Consequently, additional convergence studies are expected to yield only marginal improvements in uncertainty. In contrast, new cross-functional calculations, direct electro-optic tensor calculations, and improved calibration data would reduce uncertainty more substantially. These uses make uncertainty quantification an effective tool for guiding further computational and experimental effort.

A. Applicability, Assumptions, Limitations

Posterior certification is most valuable when uncertainty influences scientific or applicability decisions. The $BaTiO_3$ study presented here highlights two particularly important situations: the aggregation of heterogeneous datasets, and the propagation of uncertainty through nonlinear physical models. In the former, provenance-aware fusion revealed phase and tensor-component inconsistencies that would have remained hidden with a conventional literature average. In the latter, the inverse-square dependence of r_{33} on the soft-mode frequency amplified modest uncertainties in $\omega_{\rm soft}$ into substantially larger uncertainties in the derived electro-optic response. Similar challenges arise throughout computational materials science whenever results from multiple sources must be combined, such as machine learned or fitted multiscale approaches [18], or when derived quantities depend sensitively on underlying DFT observables.

The method rests on several assumptions and approximations whose impact should be understood before the certificates are used in downstream applications. These include the assumption of independent uncertainty contributions, the scope of modeled uncertainty, the availability of benchmark data for estimating methodological error, and the recursive nature of surrogate model uncertainty.

The independence assumption underlying the variance decomposition (Equation 6) is the most significant approximation. As shown in Section V C, a two-functional (LDA/PBE) covariance estimate reveals that the cross-functional shifts in Z^*, ε^{∞}, and $\omega_{\rm soft}$ partially cancel in r_{33}, making independence conservative for this observable by a substantial factor. However, this estimate is based on only two functionals and assumes the cancellation pattern generalizes across materials. A more rigorous assessment would require convergence studies that track all three quantities simultaneously, allowing covariance data to be determined directly rather than inferred from cross-functional trends. The covariance terms in Equation 19 provide the formal mechanism for incorporating such data once available.

There is, however, an opposing anti-conservative direction. The library used for the fused electro-optic posterior is dominated by semilocal functionals, so cross-functional spread within that family may understate discrepancies that would appear when comparing against hybrid, meta-GGA, quasiparticle, or experimental reference data. The band-gap case study in Section 3.2 provides the clearest example of this broader functional-family spread. The net sign of conservatism in the r_{33} certificate is therefore not established: the independence assumption likely overestimates one methodological component, while the semilocal exchangeability class may underestimate broader model discrepancy.

The framework quantifies only sources of uncertainty explicitly represented in the certificate. Unmodeled effects such as code errors, incorrect inputs, or missing physics are not captured by the four-component decomposition. However, deviations between single-calculation certificates and aggregated literature data can provide useful diagnostic information. In the present work, the discrepancy between the single-calculation and library-fused r_{33} highlighted a metadata labeling error in the reference data: The addition of new data increased rather than decreased the uncertainty in ε_{∞}, indicating that physically distinct quantities were being combined within the same average. The issue was resolved manually. The certificate framework does not diagnose such errors automatically, but it creates the conditions under which they become visible.

Similarly, the method helps identify inconsistencies in aggregated datasets, for example when quantities from different crystallographic phases or tensor conventions are combined inappropriately. In the present case, identifying the source of discrepancy required human inspection of the metadata. However, the phase, tensor-component and functional provenance tags can also provide the information needed for automated detection. Without explicit provenance tracking, phase and component conflations would be invisible within a simple literature average.

Accurate estimation of $\sigma_{\rm method}$ and $\sigma_{\rm DFT}$ requires benchmark data. For novel properties or material classes where such data is unavailable, the method must fall back on cross-functional spread as a proxy, which provides a lower bound on the true methodological uncertainty.

Finally, full convergence studies multiply computational cost, and our surrogate models for estimating $\sigma_{\rm conv}$ from single calculations introduce their own uncertainty. In principle, the framework is recursive: the surrogates themselves are candidates for certification, and their fitting error should propagate as a second-order contribution into every certificate they generate. This meta-uncertainty is not included in the present certificate and therefore marks a boundary on the reported uncertainty budget. The trade-off between certification fidelity and computational budget is application-dependent.

B. Scope of the Current Certificate

The present certificates describe bulk tetragonal $BaTiO_3$ and quantify uncertainty within the physical and computational assumptions of a calibrated soft-mode model. Important sources of electro-optic enhancement in devices, including epitaxial strain, domain engineering, finite-temperature effects, and phase stability, are outside the exchangeability class of the present certificate. Full *ab initio* electro-optic tensors including all ionic, electronic, piezoelectric, strain, and domain contributions are not represented here; the soft-mode surrogate is used precisely because it makes the bulk uncertainty chain traceable at tractable cost. The binding limitation in the present certificate is model form and descriptor selection rather than numerical convergence.

Thin-film measurements illustrate the scale of the excluded physics. Reynaud et al. [57] report an effective Pockels coefficient exceeding 550 pm/V in 150 nm epitaxial films on Si, roughly $\sim 5\times$ above the bulk value, and Suceava et al. [58] report $r_{33} = 2516 \pm 100$ pm/V at 5 K in strain-stabilized metastable films. These values are not contradictions of the bulk certificate: they belong to different physical scopes in which domain structure, strain coupling, temperature dependence, and phase stability are active variables. The $\sim$22% floor reported here is therefore irreducible only within the carried-discrepancy, semilocal-DFT soft-mode approximation used for the bulk reference condition.

C. Extension and Adoption

The certified *bulk* r_{33} is one stage in a broader uncertainty chain. For device design, performance metrics such as the effective electro-optic coefficient r_{eff} and the half-wave voltage–length product $V_\pi L = \frac{\lambda d}{2n^3 r_{\mathrm{eff}}\Gamma}$ (where λ is the wavelength, d is the electrode gap, n is the refractive index, and Γ is the optical–RF field overlap [64–66]) are obtained through additional modeling stages that introduce their own uncertainty sources. Table III illustrates this capability using a representative propagation chain from the certified bulk BaTiO3 response ($r_{33} = 108 \pm 25$ pm/V) to derived effective material parameters and device-level metrics. Library fusion improves the starting point for this propagation by reducing the uncertainty in the bulk electro-optic response from 28% for a single calculation to approximately 24% for the fused library value, approaching but not crossing the $\sim$22% floor imposed by convergence, intrinsic-DFT and model-form uncertainty. The table is a schema demonstration: the Effective and Device rows show where additional certified stages enter, while the present paper certifies the DFT and bulk-property stages.

The structured nature of the certificate metadata also creates opportunities beyond uncertainty quantification for individual calculations. Recent work in materials informatics has highlighted the importance of uncertainty-aware machine learning, particularly when training models on heterogeneous datasets assembled from multiple computational and experimental sources [67–71]. The uncertainty estimates associated with individual certificates provide a natural mechanism for incorporating data quality into model development, whether through inverse-variance weighting, calibration analysis, or heteroscedastic learning approaches that explicitly distinguish between data uncertainty and model uncertainty. More broadly, certificate metadata provides an interface for uncertainty-aware aggregation and filtering of large materials datasets; performance claims for particular learning architectures require separate benchmark studies [72–74].

IV. CONCLUSION

We have introduced A Posteriori Error Certificates (APECs), a methodology for attaching quantitative uncertainty estimates and provenance metadata to DFT-calculated values and the properties derived from them. By decomposing uncertainty into convergence, methodological, intrinsic DFT, and model-form components, the method provides a structured description of the confidence associated with computed materials properties, enables uncertainty propagation through derived quantities, supports the consistent fusion of heterogeneous data, and assigns Green, Amber and Red quality tiers to data based on overall uncertainty.

The $BaTiO_3$ application of the APECs framework demonstrates that soft-mode amplification pushes the derived Pockels coefficient to Red tier ($\sigma_{\mathrm{total}} = 28.4\%$) even when the underlying DFT-calculated properties are Green or Amber. Phase-aware partitioning of literature data resolved inconsistencies arising from mixing cubic and tetragonal data and reduced the uncertainty in the library-fused electro-optic response to approximately 24%, bounded by a $\sim$22% floor within the present carried-discrepancy semilocal-DFT soft-mode model. Although demonstrated here for the soft-mode electro-optic response of $BaTiO_3$, the method is independent of the underlying electronic-structure method or property. More broadly, APECs provide a basis for uncertainty-aware computational materials science, allowing computed properties to be compared, combined, and used systematically in databases, multistage models and data-driven materials design.

V. METHODS: A POSTERIORI ERROR CERTIFICATES

A. Sources of Uncertainty in DFT

The output of a DFT calculation [75, 76] depends on a variety of choices, each introducing distinct sources of error. These errors can be roughly grouped in three categories: intrinsic, methodological, and convergence.

Convergence is sought in materials' DFT calculations at several points. Plane-wave DFT requires truncation of the basis set at a finite energy cutoff E_{cut}, sampling of the Brillouin zone on a discrete k-point mesh, and iterative solution of the Kohn–Sham equations to a finite self-consistency threshold [63, 77, 78]. These parameters must be converged for each property and material system, and the degree to which a given calculation has achieved convergence is often not reported in online data.

At the *methodological* level, the choice of DFT code, exchange-correlation functional, basis set and pseudopotential, all introduce systematic errors that vary by property and material class [79–83]. To illustrate, we include representative bias statistics, ranging from <1% for lattice constants with the PBEsol or SCAN functionals to

TABLE III. Illustrative propagation of certified uncertainties from bulk material to device-level metrics. Each row represents a certificate-generating stage. The DFT and Property stages use the uncertainty decomposition developed in this work and summarized in Figure 3; the Effective and Device stages employ representative uncertainty estimates to illustrate how additional modeling stages can be incorporated in the same framework.

Stage	Quantity	Value	Rel. uncertainty
DFT	ω_{soft}	172 cm^{-1}	10%
↓ (×2 amplification)			
Property (single calc)	r_{33} (zz)	94 pm/V	28%
Property (library-fused)	r_{33} (zz)	94 pm/V	24%
Schematic downstream stages:			
↓ (microstructure averaging)			
Effective	r_{eff}	∼40 pm/V	∼26%
↓ (device model)			
Device	$V_\pi L$	∼2.5 V·cm	∼27%

up to ∼40% errors for band gaps using semilocal functionals, in Table S1.

The component denoted σ_{DFT} refers to residual approximate-DFT model discrepancy: uncertainty shared by practical Kohn–Sham DFT workflows after numerical convergence and known recipe-level biases have been accounted for. It includes missing or imperfectly treated physics in the chosen workflow, such as self-interaction, static correlation [84, 85], long-range dispersion when not considered explicitly, band-gap derivative-discontinuity effects [86, 87], spin–orbit coupling in heavy elements, and finite-temperature effects [88–91]. This is not a claim of an error in formally exact ground-state DFT; it is an irreducible floor for the approximate DFT workflow used in a given certificate.

The certificate framework was presented conceptually in the preceding sections. Below, we introduce mathematically each uncertainty component and the estimation procedures used to build APECs.

B. Certificate Structure

A posterior error certificate $\mathcal{C}[P]$ attaches an uncertainty estimate to a computed property $\hat{P}$ as a metadata object. It is "posterior" in the sense that it can be evaluated after the calculation, using the computational output, convergence behavior, and available benchmark data. Formally:

$$\mathcal{C}[P] = \left\{\hat{P},\, \sigma_{\mathrm{conv}},\, \sigma_{\mathrm{method}},\, \sigma_{\mathrm{DFT}},\, \sigma_{\mathrm{model}},\, \sigma_{\mathrm{total}},\, \tau,\, \Pi\right\} \tag{5}$$

where four uncertainty components are included for convergence (σ_{conv}), methodology (σ_{method}), intrinsic DFT error (σ_{DFT}), and, for derived quantities, model-form error (σ_{model}). The uncertainty components are assumed to be independent in the first approximation; details on considering correlations are provided in Section S1.1. The total uncertainty is taken as the quadrature sum (see Equation 4). The certificate also includes the quality tier estimate $\tau \in \{\mathrm{red}, \mathrm{amber}, \mathrm{green}\}$; and provenance metadata Π recording the code, version, functional, pseudopotential, all convergence parameters, the crystallographic phase and tensor convention used in the calculation. Tracking this information specifically is essential when combining literature data, since quantities such as Born effective charges and dielectric tensor can differ substantially between cubic and tetragonal phases and between zz and isotropic tensor averages.

For properties computed directly from DFT, $\sigma_{\mathrm{model}} = 0$. For derived properties (e.g., the soft-mode expression for r_{33}), σ_{model} captures the approximation error of that model and is nonzero. Implementation details are provided in Sections S2.2–S2.3 and the support repository documentation.

C. Error Model and Uncertainty Decomposition

We model the computed or derived property as [92]

$$P_{\mathrm{computed}} = P_{\mathrm{true}} + \epsilon_{\mathrm{conv}} + \epsilon_{\mathrm{method}} + \epsilon_{\mathrm{DFT}} + \epsilon_{\mathrm{model}} \tag{6}$$

The four error terms are assumed to be independent. This is the central approximation of the method and deserves explicit justification. The convergence error ϵ_{conv} is dominated by basis set truncation and k-point sampling – numerical parameters that are largely functional-independent. The methodological error $\epsilon_{\mathrm{method}}$ reflects the choice of exchange-correlation functional, pseudopotential type, and related recipe decisions. The intrinsic DFT error ϵ_{DFT} captures limitations shared by all standard functionals (e.g., self-interaction, static correlation).

In practice, correlations exist: a functional that overbinds will systematically stiffen phonons, coupling $\epsilon_{\mathrm{method}}$ to ϵ_{DFT}. We proceed with the independence assumption as a zeroth-order baseline but can estimate its impact from existing data. Comparing LDA and PBE values in the compiled literature library (Table S6), the three quantities entering the soft-mode expression for r_{33} shift systematically with the choice of functional. PBE gives slightly larger Born effective charges (+5%) and dielectric constants (+8%) than LDA, while the soft-mode frequency changes comparatively little. Because contributions from Z^* and ε^∞ enter $r_{33} \propto Z^{*2}/(\varepsilon\omega^2)$ with opposite signs, these shifts partially cancel, reducing the net variation in the derived property: the net cross-functional shift in r_{33} is only ∼2%, compared to the ∼13% expected from the independent quadrature of the individual uncertainties.

This confirms that the independence assumption is conservative for r_{33}, since it overestimates the methodological component of the r_{33} uncertainty. We retain the conservative estimate as the default because (i) the present comparison is based on only two functionals, exaggerating the cancellation effect and (ii) this cancellation is specific to r_{33}. Other quantities may instead show reinforcing correlations. When sufficient material-specific data is available, covariance terms can be included specifically. Alternatively, a more reliable estimate of the covariance could be obtained by systematically varying numerical parameters while tracking the resulting changes in ω_{soft}, Z^*, and ε^∞ (see Section S1.1).

1. *Convergence Uncertainty (σ_{conv})*

Convergence error arises from finite computational parameters [93]. For plane-wave DFT, the three dominant contributions to this are (i) basis set truncation, which decays exponentially with cutoff energy,

$$\epsilon_{E_{\text{cut}}}(E) = A_E \exp\left(-E/E_0\right); \tag{7}$$

(ii) Brillouin zone integration, which follows a power law in the number of k-points [39, 94],

$$\epsilon_k(N_k) = B/N_k^\alpha, \tag{8}$$

where $\alpha \approx 1$ for metals and $\alpha \approx 2$ for insulators; and (iii) incomplete self-consistency, arising from residual changes in the electron density at the end of the SCF cycle.

The total convergence uncertainty is the quadrature sum of all contributions [92]:

$$\sigma^2_{\text{conv}} = \sigma^2_{E_{\text{cut}}} + \sigma^2_k + \sigma^2_{\text{SCF}} + \sigma^2_{\text{FFT}} + \cdots \tag{9}$$

where additional terms such as FFT grid density, augmentation charge cutoff, Fermi smearing width, etc., are included when relevant. Details of the individual uncertainty estimates are provided in the Supporting Information.

2. *Methodological Uncertainty (σ_{method})*

Different computational "recipes", a specific combination of functional, pseudopotential [95, 96], and code, give systematically different results [97]. We model each recipe as producing an estimate biased by β_{recipe}:

$$P_{\text{recipe}} = P_{\text{true}} + \beta_{\text{recipe}} + \epsilon \tag{10}$$

When reference data exists from experiment or high-level theory (CCSD(T), DMC), biases can be estimated empirically:

$$\hat{\beta}_{\text{xc}} = \frac{1}{N_{\text{ref}}} \sum_{i=1}^{N_{\text{ref}}} \left(P_{\text{xc},i} - P_{\text{ref},i}\right) \tag{11}$$

For new chemistry, new materials or properties without benchmarks, the methodological uncertainty is estimated from the spread across available methods:

$$\sigma^2_{\text{method}} \approx \text{Var}_{\text{recipes}}[P_{\text{recipe}}] = \frac{1}{M-1} \sum_{j=1}^{M} (P_j - \bar{P})^2 \tag{12}$$

3. *Intrinsic DFT Uncertainty (σ_{DFT})*

We estimate σ_{DFT} from the residual spread relative to reference data after correcting systematic biases: [5, 6]:

$$\sigma^2_{\text{DFT}} = \frac{1}{N_{\text{ref}}} \sum_i \left(P_{\text{xc},i} - \hat{\beta}_{\text{xc}} - P_{\text{ref},i}\right)^2 \tag{13}$$

4. *Model-Form Uncertainty (σ_{model})*

For quantities derived from DFT-computed inputs, the model relating those quantities to the final observable introduces an additional uncertainty contribution, denoted σ_{model}. Here, this contribution applies to the soft-mode expression used to estimate r_{33}. We estimate σ_{model} from comparisons between the soft-mode model and DFT electro-optic calculations of the same quantity where available. For directly computed DFT properties, $\sigma_{\text{model}} = 0$ and the certificate reduces to the three-component form. For derived properties, σ_{model} may include both the model approximation error and statistical uncertainty in the calibration parameters that anchor the model to reference data.

More generally, if a surrogate model $g(\theta)$ is used in place of a higher-fidelity reference calculation or measurement F, the model discrepancy is

$$\delta = F - g(\theta). \tag{14}$$

The certificate records whether this discrepancy is *referenced* from the literature or *computed* within the same workflow. In the present $BaTiO_3$ certificate, the discrepancy is referenced: the soft-mode model-form term is carried as an uncertainty contribution rather than used to correct the central value. If an independent discrepancy estimate is available for the same exchangeability class, the same schema can represent that information as a correction with a residual calibration uncertainty.

The magnitude of σ_{model} is specific to the model and the material class. Veithen et al. computed full *ab initio* electro-optic tensors, including electronic, all ionic

TABLE IV. Quality certification tier definitions.

Tier	$\sigma_{\text{total}}/\lvert P\rvert$	Appropriate use
Green	$< 5\%$	Quantitative prediction
Amber	5–15%	Semi-quantitative, weighted fusion
Red	$\geq 15\%$	Screening, trends, ranking

modes, and piezoelectric contributions, for three perovskite oxides [10, 42]. In $BaTiO_3$ and $LiNbO_3$, the soft mode dominates the ionic response, contributing ∼80–85% of the total ionic r_{33}; the missing non-soft-mode fraction is therefore on the order of ∼15–20%. In $PbTiO_3$, by contrast, the soft mode plays only a minor role and electronic contributions dominate, making the single-mode approximation qualitatively inadequate. Paoletta and Demkov [98] independently confirmed the mode decomposition for rhombohedral $BaTiO_3$, identifying the same dominant soft mode. We adopt $\sigma_{\text{model}} = 15\%$ for $BaTiO_3$ as the nominal value within this referenced, material-specific range. For materials where the soft-mode fraction is unknown, a substantially larger σ_{model} ($\geq 50\%$) is prudent until the importance of the soft mode is established.

D. Combining and Interpreting Uncertainty Estimates

1. *Quality Certification Tiers*

We define discrete quality tiers based on total relative uncertainty, as summarized in Table IV. Green-tier certificates ($\sigma_{\text{total}}/|P| < 5\%$) are appropriate for quantitative prediction. Amber certificates (5–15%) are suitable for semi-quantitative analysis and weighted data fusion. Red certificates ($\geq 15\%$) remain useful for screening, trend identification, and ranking.

The tier thresholds are motivated by the typical uncertainty ranges across different classes of DFT-calculated properties. Structural quantities such as lattice constants and atomic positions routinely achieve <1–2% agreement across functionals and convergence settings [22], placing them comfortably within the Green tier. Response properties, including dielectric tensors and phonon frequencies typically show 5–15% cross-functional variations [11, 54], corresponding to the Amber tier. Derived quantities involving amplification or approximate physical models generally exceed 15%. These thresholds are defaults, not universal prescriptions; domain-specific applications may adjust them. In the present $BaTiO_3$ case study the resulting uncertainties separate naturally into low-uncertainty structural properties, intermediate response properties, and high-uncertainty derived electro-optic quantities.

2. *Combining Uncertainty Estimates from Various Calculations*

The APECs framework allows a principled combination of results from distinct calculations and literature data. The simplest approach uses inverse-variance weighting, while more sophisticated schemes can incorporate either known systematic biases or statistically-estimated, functional-dependent offsets of uncertain magnitude.

Inverse-Variance Weighting. The scheme weighs each property estimate according to its uncertainty, so that more precise calculations are weighted more heavily in the final result. Specifically, for N independent estimates $\{P_i \pm \sigma_i\}$ of the same quantity, the minimum-variance unbiased estimator (considering errors to be Gaussian and independent) is [99]

$$\hat{P}_{\text{fused}} = \frac{\sum_i w_i P_i}{\sum_i w_i}, \quad w_i = \frac{1}{\sigma_i^2} \tag{15}$$

with combined (fused) uncertainty

$$\sigma^2_{\text{fused}} = \frac{1}{\sum_i w_i} \tag{16}$$

The approach is particularly useful when the uncertainties are mostly independent and no large systematic biases are expected between methods. This is the methodology used in the $BaTiO_3$ case study presented in this work.

Bias-Corrected Combination. When systematic biases β_i associated with each method are known (as shown for example in Table S1), the estimates can be corrected before weighting:

$$\hat{P}_{\text{corrected}} = \frac{\sum_i w_i (P_i - \beta_i)}{\sum_i w_i} \tag{17}$$

The uncertainty now incorporates bias estimation error:

$$\sigma^2_{\text{corrected}} = \frac{1}{\sum_i w_i} + \frac{\sum_i w_i^2 \sigma^2_{\beta_i}}{(\sum_i w_i)^2} \tag{18}$$

Hierarchical Bayesian Combination. When systematic biases are uncertain, uncertainties are treated statistically, rather than explicitly corrected. The APECs framework supports hierarchical Bayesian fusion [99, 100], which can treat functional-dependent biases statistically rather than correcting them explicitly. The full model is presented in Section S1.1.

E. Uncertainty Propagation

Derived quantities Q computed from their input quantities $P_1, P_2, \ldots, P_n$ through a relationship of the form

$Q = f(P_1, P_2, \ldots, P_n)$ inherit uncertainty from those input quantities. The method propagates these uncertainties through their respective functional relationships, allowing uncertainty estimates to be assembled systematically into the derived variables.

Linear Propagation. For differentiable relationships f with small relative uncertainties, standard linear error propagation gives [92]

$$\sigma_Q^2 = \sum_i \left(\frac{\partial f}{\partial P_i}\right)^2 \sigma_{P_i}^2 + 2\sum_{i<j} \frac{\partial f}{\partial P_i}\frac{\partial f}{\partial P_j}\mathrm{Cov}(P_i, P_j) \quad (19)$$

where the covariance terms vanish if inputs are independent, as is often assumed.

Sensitivity Amplification. Some physical relationships amplify uncertainty. If $Q = P^\gamma$, then

$$\frac{\sigma_Q}{Q} = |\gamma| \cdot \frac{\sigma_P}{P}. \quad (20)$$

For soft-mode electro-optics, $\gamma = -2$ and the uncertainty is doubled. This amplification reflects the shallow energy surface associated with the soft mode: small changes in the underlying structural calculation can cause large changes in the soft mode frequency, which then propagates strongly into the derived electro-optic response. The certificate framework makes this amplification explicit rather than leaving it as an undiagnosed source of disagreement between calculations.

Beyond Linear Propagation. For nonlinear relationships f or large uncertainties where the linear approximation breaks down, Monte Carlo sampling can be used instead of linear propagation. Similarly, in multi-stage computational workflows uncertainty can be propagated sequentially through the calculation steps. This distinction matters near structural instabilities: as $\omega_{\mathrm{soft}} \to 0$, both r_{33} and its relative uncertainty become ill-conditioned, and relative Green/Amber/Red tiering ceases to be a sufficient summary. Because the present $BaTiO_3$ certificate remains in the linear uncertainty regime, linear propagation is used throughout the Results section. Monte Carlo propagation and chain propagation are described as framework capabilities in Sections S1.3–S1.4.

F. Practical Estimation of Convergence Uncertainty

Full convergence studies, involving systematic variation of cutoff energies, k-point meshes, and related numerical parameters, are computationally expensive and often impractical for large-scale or retrospective analyses. In practice, convergence uncertainty will often require estimation from incomplete convergence information. We develop surrogate models to estimate σ_{conv} from single calculations, making certification practical for large-scale screening.

Transferable Convergence Models. In many cases, convergence behaviour follows similar trends among related materials and observable. In such cases, we can estimate the convergence uncertainty from benchmark calculations performed on similar systems. We then fit property-specific convergence functions to these systematic benchmark data. The convergence uncertainty associated with energy cutoff can be estimated as[93]:

$$\sigma_{E_{\mathrm{cut}}}(E; P, \mathrm{class}) = A_P^{(\mathrm{class})} \exp\left(-E/E_0^{(\mathrm{class})}\right) \quad (21)$$

where “class” groups materials by chemistry (e.g., oxides, nitrides, metals). Similarly, the residual uncertainty associated with k-point sampling follows [39, 94]

$$\sigma_k(N_k; P, \mathrm{class}) = \frac{B_P^{(\mathrm{class})}}{N_k^{\alpha_P}} \quad (22)$$

Fitted parameters for the oxide materials class are provided in Table S3. These preliminary values define the scope of the present certificate rather than universal convergence constants.

Residual-Based Indicators. Several quantities available from single calculations can provide qualitative indicators of convergence quality and can serve as diagnostic indicators. Slow SCF convergence, unusually large stress anisotropy in nominally symmetric systems, or residual forces after relaxation may indicate incomplete numerical convergence. These indicators can be used as additional diagnostic tools when explicit convergence studies are not available, as discussed in Section S1.5.

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DATA AND CODE AVAILABILITY

The APECs $BaTiO_3$ support repository is available at `https://github.com/JulienAppleby-Millette/APEC`. It contains the machine-readable posterior certificates, curated $BaTiO_3$ literature library, figure-generation scripts, calibration outputs, and SCF convergence-study artifacts needed to reproduce the analyses reported here. The primary machine-readable data files are `data/posterior_certificates.json` and `data/bto_literature_library.json`.

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